

Mathematical Methods for Finance

Module 1 – Banach, Holder and Hilbert Spaces

Lesson 5: Function spaces, Hilbert geometry, projections and operators

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3-hour lesson structure: intuition, theory, examples, exercises, and exam-style practice

Learning outcomes for Lesson 5

By the end of this lesson, students should be able to:

- ① recognize Banach spaces and understand why completeness matters;
- ② distinguish continuous, smooth, Schwartz and Holder-type spaces;
- ③ work with inner products, orthogonality and Cauchy–Schwarz;
- ④ compute projections in Euclidean spaces and interpret Hilbert projections;
- ⑤ state Bessel inequality and use orthonormal systems;
- ⑥ decide whether a linear operator is bounded and compute simple operator norms.

How today is organised

14:30–15:20

Concepts and intuition

Banach and function spaces.

15:35–16:25

Theory and examples

Hilbert geometry and projections.

16:40–17:30

Exercises

Linear operators and approximation workshop.

Intuition: Learning rule

Each new concept follows the same route: picture, definition, worked example, counterexample, exercise.

Module 1

Block 1: Banach and function spaces

From normed spaces to Banach spaces

Definition: Banach space

A Banach space is a complete normed vector space.

Intuition: Meaning

If a sequence of approximations is Cauchy in the norm, then the limiting object still belongs to the space.

Example: Examples

$\mathbb{R}^n$, ℓ^p for $1 \leq p \leq \infty$, and $C([a, b])$ with $\|\cdot\|_\infty$ are Banach spaces.

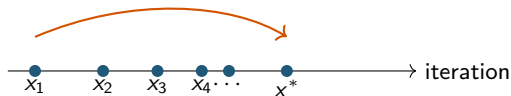
Why Banach spaces are useful

The approximation principle

Many mathematical methods produce a sequence

$$x_1, x_2, x_3, \dots$$

of approximations. If the sequence is Cauchy, completeness guarantees that the approximation process has a limit in the chosen space.



Incomplete normed space example

Counterexample: Polynomials with the uniform norm

Let P be the space of polynomials on $[0, 1]$ with $\|p\|_{\infty} = \sup_{[0,1]} |p(t)|$. Polynomials are not complete in this norm: a uniformly convergent sequence of polynomials may converge to a continuous non-polynomial function.

Example: Concrete idea

Taylor polynomials of e^t converge uniformly on $[0, 1]$ to e^t , which is not a polynomial.

Continuous, smooth and Schwartz spaces

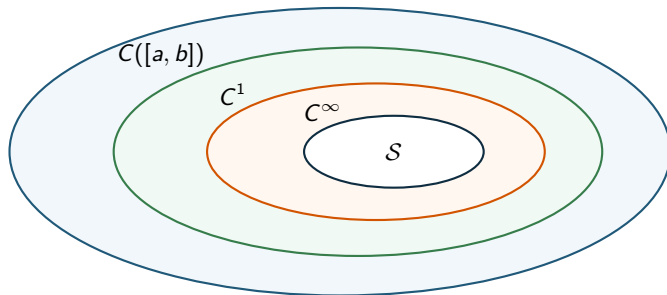
Definition: Hierarchy of regularity

- $C([a, b])$: continuous functions.
- $C^1([a, b])$: one continuous derivative.
- $C^k([a, b])$: k continuous derivatives.
- C^∞ : infinitely differentiable functions.
- $\mathcal{S}(\mathbb{R})$: Schwartz functions, smooth and rapidly decreasing with all derivatives.

Intuition: Message

The more derivatives or decay we require, the smaller and more structured the function space becomes.

Visual: spaces nested by regularity



Caution

This picture is conceptual: exact inclusions depend on the underlying domain.

Hölder spaces

Definition: Hölder continuity

A function f is Hölder continuous with exponent $\alpha \in (0, 1]$ if there exists $C > 0$ such that

$$|f(x) - f(y)| \leq C |x - y|^\alpha \quad \forall x, y.$$

Example: Special cases

- $\alpha = 1$ gives Lipschitz continuity.
- Smaller α allows rougher functions.

Exercise: Check ★★

Show that every Lipschitz function on $[0, 1]$ is Hölder continuous with exponent $\alpha < 1$.

Solution: Lipschitz implies Hölder on bounded intervals

Proof idea: Estimate

If f is Lipschitz, then $|f(x) - f(y)| \leq L|x - y|$. On $[0, 1]$, if $0 \leq |x - y| \leq 1$ and $0 < \alpha < 1$, then

$$|x - y| \leq |x - y|^\alpha.$$

Hence

$$|f(x) - f(y)| \leq L|x - y|^\alpha.$$

Thus f is Hölder continuous with exponent α .

Block 1 mini-workshop

Exercise: Exercise A ★

Explain why every finite-dimensional normed space is Banach.

Exercise: Exercise B ★★

Give a sequence of polynomials converging uniformly to a non-polynomial continuous function.

Exercise: Exercise C ★★★

For $f(x) = \sqrt{x}$ on $[0, 1]$, discuss why Lipschitz continuity fails but Hölder continuity with exponent $1/2$ holds.

Module 1

Block 2: Hilbert spaces and projections

Inner product spaces

Definition: Inner product

An inner product on a real vector space H is a map $\langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{R}$ such that

- $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0$ iff $x = 0$;
- $\langle x, y \rangle = \langle y, x \rangle$;
- it is linear in each argument in the real case.

Definition: Induced norm

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

Examples of inner products

Example: Euclidean space

In $\mathbb{R}^n$,

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i.$$

Example: Function space

On suitable functions over $[a, b]$,

$$\langle f, g \rangle = \int_a^b f(t)g(t) dt.$$

Exercise: Compute ★

Compute $\langle x, y \rangle$ for $x = (1, 2, -1)$ and $y = (3, 0, 4)$.

Solution: inner product computation

Worked example: $x = (1, 2, -1)$ and $y = (3, 0, 4)$

$$\langle x, y \rangle = 1 \cdot 3 + 2 \cdot 0 + (-1) \cdot 4 = 3 - 4 = -1.$$

Interpretation

The sign indicates whether the vectors point partly in the same or opposite directions. Orthogonality means the value is exactly zero.

Cauchy–Schwarz inequality

Theorem: Cauchy–Schwarz

In an inner product space,

$$|\langle x, y \rangle| \leq \|x\| \|y\| .$$

Proof idea: Proof idea

If $y \neq 0$, consider the nonnegative quadratic function

$$\|x - \lambda y\|^2 \geq 0.$$

Its discriminant must be nonpositive. This gives the inequality.

Orthogonality

Definition: Orthogonal vectors

Two vectors x, y are orthogonal if

$$\langle x, y \rangle = 0.$$

We write $x \perp y$.

Theorem: Pythagoras

If $x \perp y$, then

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2.$$

Exercise: Check ★

Are $(1, 1)$ and $(1, -1)$ orthogonal in $\mathbb{R}^2$?

Orthonormal systems

Definition: Orthonormal system

A family (e_i) is orthonormal if

$$\langle e_i, e_j \rangle = 0 \quad (i \neq j), \quad \|e_i\| = 1.$$

Example: Euclidean basis

The standard vectors $e_1, \dots, e_n$ form an orthonormal basis of $\mathbb{R}^n$.

Why useful

With an orthonormal system, coefficients are easy: the coefficient of x in the direction e_i is $\langle x, e_i \rangle$.

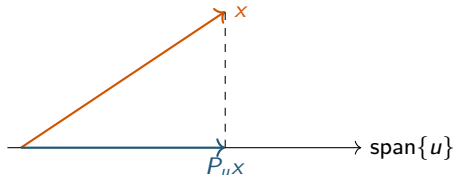
Projection onto a line

Theorem: Projection formula

If $u \neq 0$, the projection of x onto the line $\text{span}\{u\}$ is

$$P_u x = \frac{\langle x, u \rangle}{\langle u, u \rangle} u.$$

If $\|u\| = 1$, then $P_u x = \langle x, u \rangle u$.



Worked example: projection in $\mathbb{R}^2$

Worked example: Project $x = (2, 1)$ onto $u = (1, 1)$

$$\langle x, u \rangle = 2 + 1 = 3, \quad \langle u, u \rangle = 1 + 1 = 2.$$

Thus

$$P_u x = \frac{3}{2}(1, 1) = \left(\frac{3}{2}, \frac{3}{2}\right).$$

The error is

$$x - P_u x = \left(\frac{1}{2}, -\frac{1}{2}\right),$$

and it is orthogonal to u .

Hilbert spaces

Definition: Hilbert space

A Hilbert space is a complete inner product space.

Example: Examples

- $\mathbb{R}^n$ with the usual inner product.
- ℓ^2 with $\langle x, y \rangle = \sum_i x_i y_i$.
- L^2 spaces, introduced later, with $\langle f, g \rangle = \int fg$.

Message

Hilbert spaces are the natural setting for geometry and least-square approximation.

Hilbert projection theorem

Theorem: Projection theorem

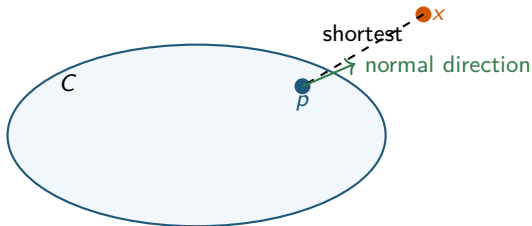
Let H be a Hilbert space and let $C \subset H$ be nonempty, closed and convex. For every $x \in H$, there exists a unique $p \in C$ such that

$$\|x - p\| = \inf_{y \in C} \|x - y\|.$$

Intuition: Meaning

There is a unique best approximation to x from the set C .

Projection theorem: geometry



Orthogonality condition

For a closed subspace M , $p = P_M x$ is characterized by

$$x - p \perp M.$$

Bessel inequality

Theorem: Bessel inequality

Let (e_k) be an orthonormal system in a Hilbert space H . Then, for every $x \in H$,

$$\sum_k |\langle x, e_k \rangle|^2 \leq \|x\|^2.$$

Intuition: Meaning

The squared lengths of the orthogonal components cannot exceed the total squared length.

Finite-dimensional picture

In $\mathbb{R}^n$ with a full orthonormal basis, Bessel becomes equality: Parseval's identity.

Block 2 mini-workshop

Exercise: Exercise A ★

Determine whether $x = (1, 2)$ and $y = (2, -1)$ are orthogonal.

Exercise: Exercise B ★★

Project $x = (3, 1)$ onto the line generated by $u = (1, 0)$.

Exercise: Exercise C ★★★

Let $e_1 = (1, 0)$, $e_2 = (0, 1)$ and $x = (a, b)$. Verify Bessel inequality and identify when equality holds.

Module 1

Block 3: linear operators and approximation

Linear operators

Definition: Linear operator

Let X, Y be vector spaces. A map $T : X \rightarrow Y$ is linear if

$$T(\alpha x + \beta y) = \alpha T(x) + \beta T(y)$$

for all $x, y \in X$ and scalars α, β .

Example: Examples

- Matrix multiplication $T(x) = Ax$.
- Differentiation $T(f) = f'$ on suitable function spaces.
- Integration $T(f) = \int_a^b f(t) dt$.

Bounded linear operators

Definition: Bounded operator

A linear operator $T : X \rightarrow Y$ between normed spaces is bounded if there exists $C \geq 0$ such that

$$\|Tx\|_Y \leq C \|x\|_X \quad \forall x \in X.$$

Definition: Operator norm

$$\|T\| = \sup_{\|x\|_X \leq 1} \|Tx\|_Y.$$

Theorem: Equivalent fact

For linear maps between normed spaces, boundedness and continuity are equivalent.

Worked example: matrix operator norm idea

Worked example: Diagonal matrix

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be $T(x_1, x_2) = (2x_1, 3x_2)$ with the Euclidean norm. Then

$$\|Tx\|_2^2 = 4x_1^2 + 9x_2^2 \leq 9(x_1^2 + x_2^2) = 9\|x\|_2^2.$$

Thus $\|Tx\|_2 \leq 3\|x\|_2$. Taking $x = (0, 1)$ gives equality, so $\|T\| = 3$.

Unbounded operator warning

Counterexample: Differentiation can be unbounded

Consider polynomials on $[0, 1]$ with the uniform norm. Let $p_n(t) = t^n$. Then $\|p_n\|_\infty = 1$, but

$$\|p'_n\|_\infty = \|nt^{n-1}\|_\infty = n.$$

Thus differentiation is not bounded from this normed space to itself.

Lesson

Linearity alone is not enough. The chosen norms matter.

Approximation as projection

General problem

Given an object x and a simpler space M , find

$$\min_{m \in M} \|x - m\| .$$

Intuition: Hilbert-space answer

If M is a closed subspace of a Hilbert space, the best approximation is the orthogonal projection $P_M x$.

Example: Practical interpretation

Approximating a complicated object by a linear combination of simple basis elements is a projection problem.

Least-squares in $\mathbb{R}^n$ as projection

Worked example: Column space

Given a matrix A , the problem

$$\min_c \|y - Ac\|_2$$

asks for the projection of y onto the column space of A . The normal equations are

$$A^T A c = A^T y.$$

Course link

This is the finite-dimensional version of Hilbert projection.

Integrated exercise 1: projection

Exercise: Problem ★★

Let $M = \text{span}\{(1, 1, 0), (0, 1, 1)\} \subset \mathbb{R}^3$. Find the projection of $x = (1, 0, 0)$ onto M .

Plan

Write $p = a(1, 1, 0) + b(0, 1, 1)$. Impose $x - p \perp (1, 1, 0)$ and $x - p \perp (0, 1, 1)$.

Solution: projection onto a plane

Proof idea: Normal equations

Let $u_1 = (1, 1, 0)$, $u_2 = (0, 1, 1)$ and $p = au_1 + bu_2 = (a, a+b, b)$. Then $x - p = (1-a, -a-b, -b)$. Orthogonality gives

$$\langle x - p, u_1 \rangle = 1 - 2a - b = 0,$$

$$\langle x - p, u_2 \rangle = -a - 2b = 0.$$

Solving, $a = 2/3$, $b = -1/3$. Hence

$$p = (2/3, 1/3, -1/3).$$

Integrated exercise 2: boundedness

Exercise: Problem ★★

Let $T : C([0, 1]) \rightarrow \mathbb{R}$ be defined by

$$T(f) = \int_0^1 f(t) dt.$$

Use the norm $\|f\|_\infty = \sup_{[0,1]} |f(t)|$. Show that T is bounded and find an upper bound for $\|T\|$.

Solution: boundedness of integration

Proof idea: Estimate

For every $f \in C([0, 1])$,

$$|T(f)| = \left| \int_0^1 f(t) dt \right| \leq \int_0^1 |f(t)| dt \leq \int_0^1 \|f\|_\infty dt = \|f\|_\infty.$$

Hence T is bounded and $\|T\| \leq 1$. Taking $f(t) = 1$ gives $T(f) = 1$ and $\|f\|_\infty = 1$, so $\|T\| = 1$.

Integrated exercise 3: finite vs infinite dimension

Exercise: Problem ★★

Explain why the closed unit ball in $\mathbb{R}^n$ is compact, but the closed unit ball in ℓ^2 is not compact.

Plan

Use Heine–Borel in $\mathbb{R}^n$. In ℓ^2 , use the unit vectors e_n and show that no subsequence is Cauchy.

Solution: finite vs infinite dimension

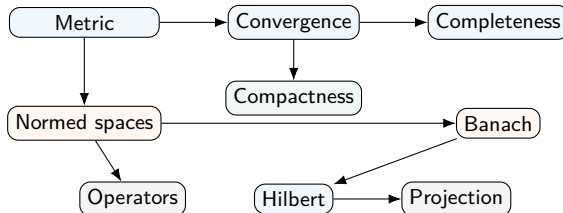
Proof idea: Comparison

In $\mathbb{R}^n$, the closed unit ball is closed and bounded; by Heine–Borel, it is compact. In ℓ^2 , the sequence (e_n) belongs to the closed unit ball. But for $m \neq n$,

$$\|e_n - e_m\|_2 = \sqrt{2}.$$

Therefore no subsequence is Cauchy, hence no subsequence converges. The closed unit ball is not compact.

Final Module 1 concept map



Module 1 in one sentence

We build spaces where limits, approximations, projections and operators can be handled rigorously.

End-of-module checklist

Theorem: Definitions

Metric, convergence, Cauchy, completeness, compactness, norm, Banach space, inner product, Hilbert space, orthogonality, projection, bounded operator.

Theorem: Theorems

Convergent implies Cauchy, closed subset of complete space, contraction theorem, Baire theorem, Heine–Borel, Holder, Minkowski, Cauchy–Schwarz, projection theorem, Bessel inequality.

Counterexample: Key counterexamples

$\mathbb{Q}$ not complete; $(0,1)$ not complete or compact; pointwise not uniform; closed bounded not compact in infinite dimension; differentiation unbounded in the wrong norm.

After-class work

Theorem: Core exercises

Exercise pack: 061, 063, 081, 083.

Example: Recommended

Exercise pack: 065, 070, 086, 092.

Advanced

Exercise pack: 078, 095, 098, 100.

Exercise: Study method

For each exercise: write the relevant definition, check all hypotheses, then compute or build a counterexample.

Mock exam: Module 1 structure

Exercise: Possible written exam structure

- ① Verify whether a formula is a metric or a norm.
- ② Study convergence/completeness/compactness in a concrete space.
- ③ Use Holder, Minkowski or Cauchy–Schwarz in an estimate.
- ④ Compute an orthogonal projection or verify boundedness of an operator.
- ⑤ Give a counterexample to a false implication.

Most important habit

Always state the definition before applying the theorem.

Bridge to Module 2

From spaces to integration

Module 1 gave us spaces, distances and norms. Module 2 uses these ideas to build measure, integration and L^p spaces.

Intuition: What changes

Functions will no longer be compared only point by point. We will also compare them through integrals and norms defined by measures.